

Control Plan - PFMEA - PFD Linkage Guide

Three-Way Linkage | AIAG | Quality Head Agents

The Three-Way Linkage

Every process step in the Process Flow Diagram (PFD) must appear in the PFMEA. Every control method in the PFMEA must be documented in the Control Plan. This three-way linkage (PFD > PFMEA > Control Plan) is a fundamental IATF 16949 requirement and a primary focus during IATF audits.

PFD to PFMEA

Process step name and number must match between PFD and PFMEA. All product and process characteristics from the PFD must appear as failure mode targets in the PFMEA. Special characteristics identified in the PFD must be marked in PFMEA.

PFMEA to Control Plan

Detection controls in PFMEA become the Control Plan inspection methods. Prevention controls in PFMEA become Work Instructions linked from the Control Plan. Special characteristics (CC/SC) flagged in PFMEA must appear in the Control Plan CC/SC column with appropriate detection method (100% or poka-yoke).

Control Plan to Work Instructions

Each control method in the Control Plan must reference a Work Instruction (WI) number. The WI describes step-by-step HOW the inspection or control is performed. This ensures operators have clear instructions, not just a column in the CP.

IATF Audit Expectation

Auditors will pick a random process step and ask you to show: (1) That step in the PFD, (2) The corresponding PFMEA row, (3) The Control Plan row, (4) The referenced Work Instruction. If any link is broken, it is a finding - typically a Minor NC for missing link, Major NC if CC/SC is not controlled.